ModelIt!

Teacher's Guide

Why Do Some Neighborhoods Flood While Others Don't?

Modeling Urban Water Systems and Environmental Justice

Quick Reference

Grade Level	Grade 11L1
NGSS Standard	HS-ESS3-4, HS-ETS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: HS-ESS3-4, HS-ETS1-1

Evaluate or refine a technological solution that reduces impacts of human activities on natural systems. Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct evidence-based explanations for disparate flood outcomes across neighborhoods and design equitable engineering solutions that reduce vulnerability in historically disinvested communities.
Disciplinary Core Idea	**ESS3.C Human Impacts on Earth Systems / ETS1.A Defining and Delimiting Engineering Problems** — Human development fundamentally alters the water cycle by replacing permeable surfaces with impervious ones. Engineering solutions must account for societal inequities and prioritize the most vulnerable populations.
Crosscutting Concept	**Cause and Effect** — Students trace the causal chain from historical policy decisions (redlining, disinvestment) through infrastructure disparities to unequal flood outcomes, identifying both proximate causes (impervious surfaces, old pipes) and root causes (systemic racism).

CAST Testing Alignment

MULTIPLE CHOICE:

Two neighborhoods receive identical rainfall during a storm. Neighborhood A has 90% impervious surfaces, 1950s storm drains, and no green space. Neighborhood B has 55% impervious surfaces, modern storm drains, and extensive tree canopy. Based on your model, approximately how much greater would you expect Neighborhood A's flood depth to be compared to Neighborhood B?

CONSTRUCTED RESPONSE:

Using evidence from your model, write a recommendation to the city council explaining why flood mitigation funding should prioritize historically disinvested neighborhoods. Include specific model data showing the infrastructure gaps and predicted flood reduction from your recommended interventions.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

When Hurricane Harvey flooded Houston in 2017, the most severely impacted areas were disproportionately low-income neighborhoods and communities of color — the same neighborhoods that had been redlined and disinvested for decades. This pattern repeats in city after city: the communities with the least political power and fewest resources face the greatest flood risk. Understanding why requires modeling both the hydrology of urban stormwater AND the history of infrastructure inequality.

Urban Hydrology: How Cities Create Floods

In a natural landscape, approximately 50% of rainfall infiltrates into the ground, 40% evapotranspires through vegetation, and only 10% becomes surface runoff. In a dense urban area, these numbers are essentially reversed: 55% becomes surface runoff, 30% infiltrates (through lawns and parks), and only 15% evapotranspires. A single acre of parking lot generates 16 times more runoff than a forested acre during the same storm. Cities literally create flooding by replacing permeable earth with impervious surfaces.

The Infrastructure Gap: Why Pipes Matter

Storm drains are designed to handle a specific rainfall intensity, typically a '10-year storm' — the statistical rainfall level expected once per decade. But infrastructure in disinvested neighborhoods is often 50-100 years old, designed for lower rainfall intensities and smaller populations. Meanwhile, climate change is increasing the frequency of extreme rainfall events. The result: storms that overwhelm aging pipes in poor neighborhoods while modern systems in affluent areas handle the same rain easily. Replacing aging storm drains costs $1-5 million per mile of pipe.

Redlining to Flooding: The Environmental Justice Connection

In the 1930s, the Home Owners' Loan Corporation (HOLC) graded neighborhoods A through D based on perceived investment risk — with D ('hazardous') neighborhoods almost always being communities of color. Redlined neighborhoods received less mortgage lending, less commercial investment, and critically, less public infrastructure spending for decades. A 2020 study in 142 US cities found that formerly redlined neighborhoods have significantly less tree canopy, more impervious surfaces, higher temperatures, and worse drainage than neighborhoods that were graded A. The flood map of today is the redlining map of 1935.

Green Infrastructure: Nature-Based Solutions

Green infrastructure uses vegetation, soil, and natural processes to manage stormwater at its source. A single mature tree intercepts 500-760 gallons of rainfall per year. A rain garden (a shallow planted depression) can absorb 30% more water than a conventional lawn. Bioswales (vegetated channels) slow and filter runoff while reducing volume by 40-60%. Green roofs absorb 50-80% of rainfall. When distributed across a neighborhood, green infrastructure can reduce stormwater runoff by 20-40%, taking pressure off aging pipe systems. Philadelphia's Green City Clean Waters program has invested $2.4 billion in green infrastructure since 2011, avoiding $8 billion in pipe upgrades.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify impervious surface coverage, storm drain capacity, green infrastructure area, runoff volume, and flood depth as the key components of the urban flood system.

E **Establish Relationships**

Students determine that impervious surfaces, storm drain capacity, and green infrastructure are external factors shaped by policy and investment decisions, while runoff volume and flood depth are internal physical outcomes determined by the interaction of rainfall with built infrastructure.

V **Visualize & Model**

Students build a computational model comparing flood outcomes in two neighborhoods with different infrastructure profiles, revealing how the same storm produces dramatically different impacts.

E **Evaluate Outcomes**

Students run disinvested neighborhood, affluent neighborhood, and green infrastructure retrofit scenarios to quantify the relationship between infrastructure investment and flood equity.

R **Revise & Extend**

Students add historical redlining score, climate change intensification, or community displacement risk to explore the deeper systemic factors driving flood inequality.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Same city. Same storm. Same amount of rain. But one neighborhood is underwater and the other is dry. Why? The answer is about a lot more than weather."*

Watch For

Show side-by-side photos of flooded vs. dry neighborhoods from the same storm in the same city. Ask: What do you think is different?

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling why floods hit some communities harder than others — and it turns out the science is inseparable from the history of race and money in America."*

Watch For

Have students read objectives. Pre-teach 'impervious surface' and 'environmental justice' with visual examples.

Slide 3: Big Question

3 min

Say: *"Look at these two maps — one from 1935, one from last year's flood. Notice anything? The neighborhoods redlined 90 years ago are the same ones that flooded. Coincidence?"*

Watch For

Show students the HOLC redlining map and modern flood data for the same city. Quick-write: Why do you think these maps match?

Slide 4: LEVER Framework

2 min

Say: *"We're going to build a model that proves whether flooding is natural or engineered — and then design a solution that fixes the injustice."*

Watch For

Preview LEVER steps. Emphasize that this model is about both science AND justice.

Slide 5: Activity 1: Components

8 min

Say: *"What determines whether a neighborhood floods? Some factors are about physics and some are about policy. Let's sort them."*

Watch For

Guide sorting: Impervious Surface Coverage, Storm Drain Capacity, and Green Infrastructure are external (shaped by human decisions). Runoff Volume and Flood Depth are internal (physical outcomes).

Slide 6: Activity 2: Connections

8 min

Say: *"More concrete means more runoff. Old pipes can't handle the water. No parks or trees to absorb rain. Each factor makes flooding worse — and they all stack up in the same neighborhoods."*

Watch For

Students map connections. Highlight: these aren't independent failures — they're compounding effects of the same root cause (disinvestment).

Slide 7: Activity 3: Simulation

10 min

Say: *"Same storm, same rainfall rate. Let's hit both neighborhoods and measure the flood depth. Then let's retrofit the vulnerable one and see what changes."*

Watch For

Students predict flood depths for each neighborhood. Run simulations side by side. Then run green infrastructure retrofit. Compare all three results.

Slide 8: Discoveries

5 min

Say: *"The data is clear: flooding isn't an act of nature in cities — it's an act of policy. And policy can be changed."*

Watch For

Show real-world data connecting redlining scores to modern flood risk. Discuss: Knowing this, what is the city's moral obligation?

Slide 9: STEM Challenge

12 min

Say: *"The city has $200 million and a mandate to make flood protection equitable. Your engineering team has to decide where every dollar goes — and prove it works with model evidence."*

Watch For

Groups design flood mitigation plans allocating budget between green infrastructure, pipe upgrades, and other interventions. Present with model evidence.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Impervious Surface Coverage, Storm Drain Capacity, Green Infrastructure Area
- **INTERNAL:** Runoff Volume, Flood Depth

Relationships

- **Impervious Surface Coverage to Runoff Volume:** POSITIVE (+) — More impervious surface means less water infiltrates into the ground during a storm. At 90% coverage, nearly all rainfall becomes surface runoff; at 30% coverage, most rain soaks in. Every additional percentage point of pavement increases the flood risk for a neighborhood.
- **Storm Drain Capacity to Flood Depth:** NEGATIVE (-) — Higher storm drain capacity allows more water to be channeled underground and away from the neighborhood, reducing surface flooding. Neighborhoods with modern, high-capacity drainage can handle storms that would overwhelm older, undersized systems.
- **Green Infrastructure Area to Runoff Volume:** NEGATIVE (-) — Green infrastructure absorbs, stores, and slowly releases rainwater, reducing the total volume of runoff that must be handled by storm drains. A comprehensive network of rain gardens, bioswales, tree canopy, and permeable pavement can reduce neighborhood runoff by 20-40%.

Key Discoveries

- Neighborhoods with high impervious surface coverage and outdated storm drain infrastructure experience dramatically higher flood depths during the same storm — flood risk is determined by infrastructure, not weather alone
- Historical patterns of disinvestment created the infrastructure disparities that cause unequal flood outcomes today — neighborhoods that were redlined in the 1930s-60s consistently have worse drainage systems and less green space
- Green infrastructure can significantly reduce flood risk in vulnerable neighborhoods, but it cannot fully compensate for undersized storm drains — a comprehensive approach requires both green infrastructure AND pipe upgrades
- The most cost-effective flood mitigation strategy combines green infrastructure (which slows and absorbs runoff at the surface) with targeted storm drain upgrades (which increase underground capacity for remaining runoff)

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Flooding is a natural disaster that affects everyone equally"	While the rain itself is natural, urban flooding is largely determined by human infrastructure decisions. Neighborhoods with modern storm drains, green space, and lower impervious surface coverage experience significantly less flooding from the same storm. The model clearly shows that flood depth is a function of infrastructure quality, not just rainfall intensity. And infrastructure quality is a function of historical investment decisions that systematically favored white and wealthy communities.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Green infrastructure is just decorative landscaping"	Green infrastructure is engineered to manage stormwater — rain gardens, bioswales, permeable pavement, and tree canopy are carefully designed systems that absorb specific volumes of water at specific rates. A properly designed bioswale can reduce runoff by 40-60% along a street. Philadelphia's $2.4 billion green infrastructure program has measurably reduced combined sewer overflows across the city. These are engineered systems with quantifiable performance metrics, not just landscaping.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Just building bigger pipes will solve urban flooding"	Larger pipes can handle more water, but they are extremely expensive ($1-5 million per mile), disruptive to install, and don't address the root cause — too much water reaching the pipes in the first place. Green infrastructure reduces runoff at the source, decreasing the volume that pipes must handle. The most effective and cost-efficient approach combines green infrastructure (to reduce volume) with targeted pipe upgrades (to increase capacity for remaining runoff). Philadelphia found this combined approach saved $5.6 billion compared to a pipes-only strategy.